- Which type of electroacoustic transducer converts the electrical charges produced by some forms of solid materials into energy?

    B. Piezoelectric

- In the transfer function $G(s) = 10/(s^2 + 5s + 6)$, the system classified as:

    D. Second-order

In a magnetically coupled circuit, how does increasing air gap in the core affect the mutual inductance?

A. It decreases the mutual inductance
B. It increases the mutual inductance
C. It does not affect mutual inductance
D. It decreases the self-inductance but not the mutual inductance

Which statement is TRUE about Zener diodes?
A. Prevents current from exceeding a certain level
B. Powers forward voltage drop than standard diodes
C. Maintains a stable voltage across during operation
D. Operates in the reverse breakdown region

A MOSFET amplifier has a high-frequency roll-off due to the Miller effect. How can this be mitigated?
A. Increase the gate-source capacitance Cgs
B. Increase the transconductance gm of the MOSFET
C. Reduce the gain of the amplifier
D. Increase the drain resistance RD

A Building Management System (BMS) is implemented using AI, IoT, and data analytics. The property administrator is paying high electricity costs due to multiple equipment operating simultaneously. How can energy optimization be achieved by the BMS?

A. Monitoring and controlling key building systems
B. Scheduling of building operation
C. Predicting fault and alarm occurrence
D. Understanding energy demands and assessing their impact

A current flowing through a resistor in a circuit is 158 mA and the resistance is 1,000 Ω. What is the source voltage of the circuit?

A. 128 V
B. 438 V
C. 328 V
D. 158 V

Which power supply component is responsible for converting AC voltage to pulsating DC voltage?

A. Regulator
B. Rectifier
C. Voltage multiplier
D. Filter capacitor

Which JFET amplifier has a high input impedance, low output impedance (Zout), and a voltage gain less than 1?

A. Common-source amplifier
B. Common-emitter
C. Common-gate amplifier
D. Source follower

A feedback control system has a forward path gain G = 10 and a feedback path gain H = 2. What is the primary feedback ratio of the system?

A. 2/21
B. 10/21
C. 20/21
D. 1/21

In a counter circuit with a 5-bit binary sequence, how many flip-flops are needed to represent all possible states?

A. 4 flip-flops
B. 6 flip-flops
C. 5 flip-flops
D. 7 flip-flops

Answer: C

Your team is creating an ASM chart for a traffic light controller. Curlee places states like “Red Light” and “Green Light” inside an oval. Julius uses diamonds to indicate conditions such as “Is Timer Expired?”. Sarah places action “Turn ON Yellow Light” inside a rectangle. Which among the three followed the correct ASM chart correctly?

A. Sarah only
B. Curlee only
C. Julius only
D. Julius and Sarah

Answer: D

A capacitor of 60 μF has a voltage given by
$v(t) = 25 \times 10^3\, t$ V, for $0 < t < 2$ ms
What is the maximum stored energy Wmax?

A. 37.5 MJ
B. 150.5 mJ
C. 1.87 mJ
D. 75.0 mJ

Answer: D

A battery can supply 16 J of energy to move 10 C of charge. What is the voltage of the battery?

A. 160 V
B. 0.625 V

C. 4.33 V
D. 1.6 V

Answer: D

In a commercial building with high-security requirements, a fire alarm system must be integrated with access control systems to avoid interference.

A. Disconnecting the fire alarm from the access control system to avoid interference
B. Manually unlocking all doors upon fire alarm activation
C. Making the security doors remain unlocked during emergencies
D. Using a relay-based trigger to unlock doors during emergencies

Answer: D

Find the required collector feedback bias resistor for an emitter current of 1 mA, a 4.7K collector load resistor, and a transistor with $\beta = 100$. Find the collector voltage VC.

A. 4.5 V
B. 5.3 V
C. 3.5 V
D. 8 V

Answer: B

The feedback element used in a Wien-bridge oscillator has an input voltage of 5 V at its resonant frequency. What will be the output voltage?

A. 1.67 V
B. 2.50 V
C. 2.67 V
D. 50 V

Answer: A

A mod-8 counter requires how many flip-flops?

A. 3
B. 2
C. 4
D. 8

Answer: A

In a simple crystal radio, a capacitor–inductor (LC) circuit is connected to the antenna and ground. Since thousands of sine waves from different radio stations hit the antenna, what is the role of the oscillator?

A. Tuner
B. Rectifier
C. Amplifier
D. Filter

Answer: A

In practical application, what system instructs or tells the computer what to do when it boots up?

A. Read-Only Memory
B. Basic I/O System
C. Embedded Systems Memory
D. Random-Access Memory

Answer: B

In an access control system for a high-security building, what is the MOST effective way to prevent unauthorized access due to credential theft?

A. Apply multi-factor authentication with biometrics
B. Use a centralized database for access logs
C. Increase the encryption strength of stored credentials
D. Implement smart card access only

Answer: A

A differential amplifier implemented with two bipolar junction transistors has the following parameters: V_{CC} = 10 V, RC = 10 kΩ, RE = 10 kΩ, β = 250. If the base current is 71.5 μA, what is the approximate value of the emitter current?

A. 1.43 μA
B. 25 μA
C. 86 μA
D. 18 mA

Answer: A

Sudden voltage spikes are causing accidental triggering of the SCR. Which circuit modification would be the MOST effective in preventing this incident?

A. Adding a series capacitor across the SCR's anode and cathode
B. Placing a resistor in parallel with the gate terminal
C. Using a snubber circuit (a resistor and capacitor in series) across the SCR
D. Connecting an additional diode in reverse bias across the SCR

Answer: C

In a root locus plot, what is the result of adding a pole to the system?

A. Shifting the poles further into the left half-plane
B. Increasing the phase margin
C. Moving the root locus closer to the imaginary axis
D. Decreasing the system gain

Answer: C

When does a race condition occur in an asynchronous sequential circuit?

A. The state machine's clock is not synchronized
B. Two or more state variables change simultaneously
C. The circuit cannot decide on the next state
D. The outputs stabilize after a long delay

Answer: B

A microprocessor has a 20-bit address bus. What is the maximum addressable memory size?

A. 256 KB
B. 2.5 MB
C. 512 KB
D. 1 MB

Answer: D

In the frequency response analysis of a system, what will happen to the system if the system's phase margin is negative?

A. Marginally stable
B. Stable
C. Oscillatory
D. Unstable

Answer: D

The input frequency is 1 kHz in a series circuit with a capacitive reactance of -j200 Ω, inductive reactance of 750 Ω, and a resistor. What is the capacitor value?

A. 586 nF
B. 200 nF
C. 986 nF
D. 796 nF

Answer: D

At a specific distance from the LED source, what refers to the power received per unit area?

A. Irradiance
B. Luminous intensity
C. Luminance
D. Radiant intensity

Answer: A

You are designing a digital circuit that needs to output HIGH only when 3 specific conditions are true simultaneously. The logic must be implemented using ICs from the 74xx family. Which IC would be the MOST efficient choice to minimize the number of gates and wiring?

A. 7408
B. 7404
C. 7411
D. 7432

Answer: C

Calculate the floating-point rate for a single processor with two floating-point units per processor and a frequency F = 1.1 GHz.

A. 0.1690 TF
B. 0.0044 TF
C. 0.00528 TF
D. 0.0132 TF

Answer: B

Which statement is TRUE about a PIN Diode?

A. Low reactance in the forward direction
B. High impedance in the reverse direction
C. High reactance in the reverse direction
D. High impedance in the forward direction

Answer: B

The system is experiencing a potential interrupt conflict. How can you use the Read Interrupt Mask (RIM) instruction to determine the status of the program?

A. Report the accumulator bits that indicate interrupt mask status
B. Simulate the bits to identify possible conflicts
C. Interpret the accumulator bits to check interrupt status and mask settings
D. Clear the interrupt flags automatically

Answer: C

In the small-signal model of a diode, what component represents the resistance that changes with operating point?

A. Breakdown resistance (Rb)
B. Series resistance (Rs)
C. Junction capacitance (Cj)
D. Dynamic resistance (rd)

Answer: D

What is the utility program that translates assembly language source code into executable machine code?

A. Assembler
B. Application
C. Register
D. Instruction

Answer: A

What happens to the non-linear distortion of the output signal caused by the initiation of negative feedback?

A. Decreases
B. Increases
C. Changes sinusoidal
D. Remains stable

Answer: A

What circuit component PRIMARILY contributes to the high-frequency gain loss of a BJT amplifier?

A. The power supply ripple
B. The coupling capacitors
C. The base-emitter junction capacitance
D. The emitter resistor

Answer: C

Which condition results in the HIGHEST voltage gain (Av) in an amplifier?

A. Load resistance at its minimum value with a high source resistance
B. Load resistance at its minimum value with a low source resistance
C. Load and source resistance at their highest values
D. Load resistance at its maximum value with a low source resistance

Answer: D

What determines the frequency response of an op-amp?

A. The input impedance and overall frequency
B. The open-loop gain and gain–bandwidth product
C. The output impedance and overall frequency
D. The power supply voltages and bandwidth product

Answer: B

What type of circuit is used for overvoltage protection?

A. Thermistor
B. Crowbar
C. Suppressors
D. Clipper

Answer: B

Technician Paul says a relay is used to control a small amount of current in a circuit with a large amount of current. Technician Roi says a relay is used to create a voltage across a control circuit. Whose statement is TRUE about a relay?

A. Technician Roi
B. Neither Technician Paul nor Roi
C. Both Technicians Paul & Roi
D. Technician Paul

Answer: D

In a 3-input OR gate, what will be the output when all inputs are 0?

A. Undefined
B. 1
C. 0
D. Infinity

Answer: C

What is measured by the mutual inductance between two coils?

A. Total flux linking the coils
B. Voltage induced in one coil by a current in the other
C. Inductance of the individual coils
D. Resistance of the coils

Answer: B

An engineer needs to install a device interconnection in a network of computers. The input/output ports need to be considered to have a smooth implementation in the microcontroller. What step should the engineer follow to provide the microcontroller with the I/O signals to the devices?

A. Use dedicated I/O ports and peripherals
B. Convert analog to digital data for device processing
C. Use communication ports to devices
D. Send digital pulses to external devices

Answer: A

What is the purpose of a coupling capacitor?

A. To block high-frequency noise
B. To allow AC signals to pass while blocking DC components
C. To block low-frequency noise
D. To allow DC signals to pass while blocking AC components

Answer: B

What values of "s" make the denominator equal to zero, defining the poles of the transfer function?

A. Negative
B. One
C. Zero
D. Positive

Answer: A

A BJT small-signal amplifier has a capacitor bypassing the emitter resistor. What happens if the capacitor is removed?

A. The DC operating point changes significantly
B. The frequency response shifts to lower frequencies
C. The gain of the amplifier decreases
D. The input impedance decreases

Answer: C

A voltage multiplier circuit is designed to step up an AC voltage from 20 V RMS to a theoretical 80 V peak DC. However, the measured output is only 60 V. What is the MOST probable cause of this result?

A. The capacitors are too small, limiting charge storage
B. The input frequency is too high, preventing full charging
C. The diodes have excessive reverse leakage current
D. The load resistance is too high, causing voltage drop

Answer: A

The characteristic equation of a given system is $s^4 + 11s^2 + 6s + k = 0$. What restriction must be placed on the parameter k to ensure that the system is stable?

A. $-1 < k < 5$
B. $0 < k < 15$
C. $0 < k < 10$
D. $-3 < k < 3$

Answer: C

The operating frequency of a circuit is 500 Hz. We want to raise it by two octaves. What is the corresponding frequency?

A. 250 Hz
B. 2 kHz
C. 1 kHz
D. 4 kHz

Answer: B

What is the main PURPOSE of using feedback in an amplifier circuit?

A. To increase amplifier gain and reduce distortion
B. To increase the output impedance and reduce distortion
C. To improve bandwidth and reduce distortion
D. To increase the input resistance and reduce distortion

Answer: C

The overall voltage gain in a cascaded amplifier configuration is approximately equal to the:

A. Sum of the individual voltage gains
B. Difference of the individual voltage gains
C. Average of the individual voltage gains
D. Product of the individual voltage gains

Answer: D

What does Gauss's law for magnetism state in Maxwell's equations?

A. The net magnetic charge in a region is zero
B. The magnetic field is due to electric current only
C. The magnetic flux density through a surface is proportional to the enclosed magnetic charge
D. The electric flux through a surface is proportional to the enclosed charge

Answer: A

The admittance of a parallel circuit is $0.12 \angle -30°$ S. The circuit is:

A. Inductive
B. Reactive
C. Resistive
D. Capacitive

Answer: A

What theorem states that a circuit with multiple power sources can be analyzed by evaluating only one power source at a time?

A. Node Voltage Method
B. Thevenin's Theorem
C. Maximum Power Transfer Theorem
D. Superposition Theorem

Answer: D

When an electronic circuit replicates a current from one branch to another and acts like a current amplifier, this is called:

A. Differential Circuit
B. Source Voltage
C. Current Source
D. Current Mirrors

Answer: D

In a common-emitter amplifier, what condition must be met to ensure a high voltage gain?

A. Reduce emitter resistance while increasing collector resistance
B. Reduce collector resistance while keeping emitter resistance constant
C. Increase base resistance while reducing collector resistance
D. Increase emitter resistance while keeping collector resistance constant

Answer: A

Two inductances L1 = 10 mH and L2 = 20 mH are connected in a parallel circuit. What is the equivalent inductance?

A. 12.12 mH
B. 7.66 mH
C. 4.37 mH
D. 6.67 mH

Answer: D

When designing a BJT amplifier for high-frequency operation, the overall gain-bandwidth product is PRIMARILY influenced by what factor?

A. The collector-emitter capacitance
B. The biasing network
C. The input and output coupling capacitors
D. The emitter resistance

Answer: A

A certain 120 VAC RMS, 60 Hz, 1.2 HP motor has an efficiency of 90% and a lagging power factor of 0.85. What is the reactive power drawn from the system? (Assume: 1 HP = 745.7 W)

A. 817 VAR
B. 975 VAR
C. 617 VAR
D. 770 VAR

Answer: C

Given the set of impedance parameters, which is representative of a bipolar junction transistor operating in common-emitter configuration. The two-port is driven by an ideal sinusoidal voltage source (Vs) in series with a 50 Ω resistor and terminated in a 10 kΩ load resistor. What is the voltage gain?

A. -555
B. -450
C. 580
D. -333

Answer: D

In a D flip-flop, under what condition will the output Q be high when the input D is low?

A. The clock is low
B. The clock is high
C. The clock edge triggers the flip-flop
D. The reset signal is active

Answer: C

How does the magnetic field behave inside a conductor carrying a steady current?

A. An exponential decay
B. A linear relationship with radius
C. A constant value
D. An inverse-square law

Answer: B

Under what condition will a P-channel JFET connected to the source be biased positively?

A. Biased positively
B. Connected to the ground
C. Biased negatively
D. Zero voltage

Answer: C

What type of circuit transforms current and charge in the process of current flow?

A. Lumped
B. Tuned
C. Inductive
D. Linear

Answer: C

What type of memory has the advantage of being writable but needs to be physically removed from the circuit to be erased and reprogrammed?

A. Static RAM
B. Flash eRAM
C. Register
D. Electrically programmable ROM

Answer: D

What describes the Curie temperature of a ferromagnetic material?

A. It loses its ferromagnetic properties and becomes paramagnetic
B. Its permeability becomes infinite
C. It becomes a perfect conductor
D. It achieves maximum magnetization

Answer: A

What industrial computer control system monitors input and output devices based on a custom program?

A. Distributed Control Systems
B. Programmable Logic Controller
C. Supervisory Control and Data Acquisition
D. Remote Terminal Units

Answer: B

When will the output of a 3-input AND gate become HIGH?

A. At least one input is 1
B. At least one input is 0
C. Any two inputs are 1
D. All inputs are 1

Answer: D

What part of a computer system manages the storage and retrieval of data and instructions for the processor?

A. I/O Subsystem

B. User Interfaces
C. Memory Subsystem
D. Central Processing Unit

Answer: C

What is the PURPOSE of the source code digitalRead() function in an Arduino?

A. Reads the value from a specified digital pin, either HIGH or LOW, and returns this value
B. It can read or write signals that are HIGH or LOW
C. Configures the specified pin to behave either as INPUT or OUTPUT
D. Writes a HIGH or a LOW value to a digital pin

Answer: A

What is the use of a summing point in a block diagram representation of a control system?

A. Multiply signals
B. Integrate the signals
C. Amplify the signals
D. Add or subtract signals

Answer: D

What will happen to a charged particle that enters a magnetic field perpendicularly?

A. Moves in a straight line
B. Accelerates in the field direction
C. Follows a helical path
D. Moves in a circular path

Answer: D

Which oscillator uses a positive feedback loop for sustained oscillations?

A. Low-pass filter
B. Differential amplifier
C. Voltage-controlled oscillator
D. Hartley oscillator

Answer: D

An inverting amplifier gain is sensitive to the Miller effect. What will happen to the input impedance of the amplifier due to the influence of parasitic capacitance?

A. Decreased
B. Unaffected
C. Increased
D. Phase change

Answer: A

Which is NOT relevant to the functioning of optoelectronic devices?

A. Light Control
B. Light Detection
C. Light Generation
D. Light Simulation

Answer: D

A capacitor is connected in series with a 5 kΩ resistor and a 24 V DC supply. What is the initial rate of voltage change across the capacitor immediately after the DC supply is connected?

A. 88 V/s
B. 102 V/s
C. 67 V/s
D. 138 V/s

Answer: A

Convert the binary number 110011110101_2 to octal.

A. 1743
B. 3317
C. 6365
D. 6535

Answer: C

Three resistors of 25 Ω, 50 Ω, and 100 Ω are connected in series with a 100 V power source. What is the voltage across the 50 Ω resistor?

A. 14.3 V
B. 28.6 V
C. 50.0 V
D. 57.1 V

Answer: B

What component in Ampere's law contributes to the displacement current?

A. The time-varying electric field in a dielectric
B. Current that flows due to the magnetic field
C. The electric field that induces magnetic fields in space
D. The induced voltage due to the magnetic flux

Answer: A

A parallel AC circuit supplied with 120 Vac results in a total current of 5 A. If the phase angle is -53.13°, how much real power is dissipated by the circuit?

A. 3.6 kVA
B. 480 W
C. 600 VA
D. 360 W

Answer: D